

Intramodal Dispersion Properties of Step-Index Few-Mode Spun Fibers

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(Invited Paper)

Abstract—The intramodal dispersion within groups of quasi-degenerate modes in step-index few-mode fibers is studied with a semianalytic approach. By means of theoretical analyses based on accurate modeling of mode coupling, the most general problem of arbitrary high-order modes is simplified to five different cases, which are analyzed numerically. Results show that the spin may reduce intramodal dispersion if its rate is tuned finely, similarly to single-mode fibers. At variance with this case, however, spin may also increase the intramodal dispersion of high-order modes for particular conditions. Being limited to intramodal dispersion, the analysis reported here does not consider the modal dispersion accumulated between different groups of nondegenerate modes. Nonetheless, results find application in the so-called low mode coupling regime.

Index Terms—Few-mode fibers, modal dispersion, mode coupling, spun fibers.

I. INTRODUCTION

Spatial division multiplexing (SDM) is the new paradigm that aims at facing the foreseen capacity crunch of fiber optic communication systems [1], [2]. A strong research effort is focusing on all the technical aspects of SDM systems, including new classes of multi-mode and multi-core fibers [3]–[6]. Regarding multi-mode fibers, in particular, there is a great interest in optimizing the transmission properties by minimizing modal dispersion and mode-dependent loss [7]. So far this goal has been pursued mainly by a careful design of the fiber index profile [8], [9]. Nonetheless, there is still an open question about whether or not further improvements can be attained by spinning the fiber during the drawing so to enhance its intrinsic mode coupling [10].

Actually, spin has been widely studied in the framework of single-mode fibers, where it has proved to be the most effective way to mitigate polarization mode dispersion (PMD) [11]–[13]. Simply put, spin reduces PMD by recovering “on average” the cylindrical symmetry of the fiber [14], but the complete pic-

ture is much more involved, to the extent that, under peculiar conditions, spin may actually even increase the PMD of single-mode fibers [15]. To foresee what could be the effect of spin on multi-mode fibers is therefore not trivial and this paper aims at shedding some light on the topic.

Under the weak-guide approximation, light propagation in an optical fiber can be described by LP modes [16]. As well known, however, each LP mode is actually a group of 2 or 4 (quasi) degenerate hybrid modes; therefore, modal dispersion of a fiber has two contributions. One comes from the difference among the propagation constants of LP modes of different order; we call this contribution the *inter-modal* dispersion. The second one comes from the difference among the propagation constants of the hybrid modes that make each LP group of degenerate modes; this is what we call the *intra-modal* dispersion. In multi-mode fibers the inter-modal dispersion is much larger than the intra-modal one, which would be zero if the LP groups were exactly degenerate.

Despite the apparently marginal role of intra-modal dispersion, in this paper we study the effects that spin has on it, neglecting inter-modal dispersion. The main reason for this choice is practical. At the time being a theory describing the properties of multi-mode spun fibers is completely missing. Furthermore, the parameter space of a multi-mode spun fiber would be so wide to explore that a brute-force numerical approach to the problem seems not viable. Differently, intra-modal dispersion is a simpler subset of the general problem whose analysis may help shedding more light on the complete picture.

Most important, however, is the fact that whether or not the intra-modal dispersion could be neglected with respect to the inter-modal one depends on the way the SDM transmission is implemented. Actually, one of the possible approaches to SDM consists in minimizing as much as possible the mode coupling among LP modes, so that each group of quasi degenerate modes can be used as an independent transmission channel, simplifying (if not avoiding altogether) the MIMO equalizer at the receiver [6], [17]–[20]. In this transmission scheme the inter-modal dispersion has no effect, whereas the residual intra-modal dispersion plays the same role that PMD plays in single-mode fibers. In this perspective, the analysis of intra-modal dispersion in spun fibers can help in designing fibers with better performances.

Following the above argumentation, we therefore focus the attention on the effects that spin has on intra-modal dispersion of few- and multi-mode step-index fibers. After a general theoretical introduction in Section II, we propose an accurate theoretical model to describe the intrinsic perturbation affecting a fiber in

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Section III. In particular, we highlight that in the present case only the intrinsic sources of coupling should be considered, and this limits the analysis to geometrical deformations of core and/or cladding, and stress birefringence induced by the interaction between the geometrical deformations themselves and the effect of thermal expansion. Section IV specializes the analysis further to spun fibers, leading to the identification of five main cases. Finally, in Section V we numerically study these cases, showing that they are special conditions in which the spin may actually increase the intra-modal dispersion of four-fold degenerate LP modes. These conditions apart, however, spin decreases intra-modal dispersion, despite its effectiveness might be not as high as in single-mode fibers.

II. GENERAL THEORY

As customary, we describe light propagation along a multi-mode fiber by the generalized $N \times N$ complex Jones matrix $\mathbf{U}(z)$, where z is the longitudinal coordinate along the fiber, and N is the number of propagating modes (we count each degenerate mode independently, so for single-mode fiber $N = 2$) [21]. Neglecting mode-dependent loss (as will be always the case in this paper), the variation of $\mathbf{U}$ with z depends on the mode coupling properties of the fiber and we can write [22]:

$$\frac{\partial \mathbf{U}}{\partial z} = -j\mathbf{B}(z)\mathbf{U}(z) = -j[\mathbf{D} + \mathbf{K}(z)]\mathbf{U}(z), \quad (1)$$

where $\mathbf{D} = \text{diag}(\beta_1, \dots, \beta_N)$ is a diagonal matrix (usually independent of z), β_i is the propagation constant of the i th mode, and $\mathbf{K}(z)$ is a Hermitian matrix representing mode coupling conditions at position z along the fiber. As described in Section III, the elements of $\mathbf{K}$ can be calculated with the coupled-mode theory for each kind of perturbation acting on the fiber [23], [24].

In single-mode fibers ($N = 2$) we are used to describe the strength of the perturbations in terms of birefringence and beat length [25]. It is convenient to introduce analogous parameters also for the most general case. Specifically, we define the *coupling strength*, $\Delta\kappa$, as the difference between the maximum and the minimum eigenvalues of $\mathbf{K}$:

$$\Delta\kappa = \max\{\text{eig } \mathbf{K}\} - \min\{\text{eig } \mathbf{K}\}; \quad (2)$$

similarly, we define the *coupling beat length* as $L_C = 2\pi/\Delta\kappa$. Both $\Delta\kappa$ and L_C depend in general on z and describe the local coupling characteristics of the fiber. For $N = 2$ they coincide with the classical definition of birefringence strength and beat length, respectively.

Differently from single-mode fibers, multi-mode fibers are affected also by intrinsic difference in mode propagation constants. To quantify this aspect, similarly to what done for the mode coupling, we use the *modal birefringence* [26], $\Delta\beta$, as the difference between the maximum and minimum eigenvalues of $\mathbf{D}$, and the *modal beat length* as $L_M = 2\pi/\Delta\beta$. Note that $\Delta\beta$ is simply the difference between the maximum and the minimum propagation constants and it is zero only for groups of degenerate modes.

The modal dispersion properties of the fiber at a given distance z are related to the dependence of $\mathbf{U}$ on the optical frequency

ω . In general we can write

$$\frac{\partial \mathbf{U}}{\partial \omega} = -j\mathbf{Q}(z, \omega)\mathbf{U}(z, \omega), \quad (3)$$

where the Hermitian matrix $\mathbf{Q}$ contains all the information about the modal dispersion of the fiber. In fact, the N eigenvalues τ_n of $\mathbf{Q}$ are the group delays of the so called principal modes of propagation, which are represented by the corresponding eigenvectors [21]. It is customary to summarize the dispersion properties of the fiber by two parameters [21], [22], [27], [28]. The first one is the *differential group delay* (DGD) or *maximum delay spread*, $\Delta\tau$, which is the difference between the maximum and the minimum group delay: $\Delta\tau = \max_n\{\tau_n\} - \min_n\{\tau_n\}$. This parameter has a direct impact on SDM system design, for the larger the DGD the longer the equalizing filter at the MIMO receiver. The second parameter is the *root mean square modal dispersion* (rms-MD) defined as $\tau = (N \sum_n \tau_n^2)^{1/2}$. For $N = 2$, $\Delta\tau$ and τ coincide, but in general the relationship between them is not trivial and the impact of τ on the SDM system complexity is less direct. Nonetheless, τ offers a theoretical advantage because it can be “easily” evaluated as $\tau^2 = N \text{tr } \mathbf{Q}^2$, where $\text{tr } \mathbf{A}$ is the trace of matrix $\mathbf{A}$.

To quantify the effects of spin on the modal dispersion, we evaluate the spin induced reduction factor (SIRF), defined as the ratio between the mean DGD of the fiber and the mean DGD that the same fiber would have if it were not spun:

$$\text{SIRF} = \frac{\langle \Delta\tau \rangle}{\langle \Delta\tau_{\text{unspun}} \rangle}. \quad (4)$$

As clarified later, the mean values are evaluated with respect to the statistical ensemble of the random perturbations. Similarly, we evaluate the SIRF also with respect to the rms-MD as:

$$\text{SIRF}_{\text{rms}} = \sqrt{\frac{\langle \tau^2 \rangle}{\langle \tau_{\text{unspun}}^2 \rangle}}. \quad (5)$$

As for τ , the interest in this version of the SIRF is due to mathematical convenience, since it should be easier to analytically evaluate $\langle \tau^2 \rangle$ rather than $\langle \Delta\tau \rangle$.

Mode coupling and modal dispersion are linked by a cause-effect relationship, described by the dynamical equation [22]

$$\frac{\partial \mathbf{Q}}{\partial z} = \frac{\partial \mathbf{B}}{\partial \omega} + j(\mathbf{B}\mathbf{Q} - \mathbf{Q}\mathbf{B}), \quad (6)$$

which shows that modal dispersion depends not just on $\mathbf{B}$ but also on its ω -derivative, $\mathbf{B}_\omega$. Note that in general $\mathbf{B}$ and $\mathbf{B}_\omega$ are independent of each other, and depend on the specific propagation constants and coupling conditions of the fiber. In the literature about PMD it is customary to assume that $\mathbf{B}_\omega = \mathbf{B}/\omega$ [13], [29], [30]; accordingly we make the same assumption here. However, we perform numerical simulations also considering the exact expression of $\mathbf{B}_\omega$, and we will show that the two models agree, except for a range of values of marginal practical interest.

III. INTRINSIC MODE COUPLING

Not all the possible mathematical expressions of the complex coupling matrix $\mathbf{K}$ are physically sound. Spin is intended to

mitigate the effects of intrinsic mode coupling; in fact, the intrinsic sources of mode coupling are only geometrical deformations of core and/or cladding, and stress birefringence induced by the interaction between the geometrical deformations themselves and the effect of thermal expansion [31]. Hence, the effect of spin should be studied only with respect to these perturbations. Note that this clarification was not necessary in the analysis of single-mode fibers, where all sources of perturbation (excluding twist and Faraday rotation) give raise to linear birefringence (i.e., to the same coupling matrix) [25], [32].

In this section we recall the main results about stress birefringence and geometric deformations. Coupling coefficients may have rather complex analytic expressions, therefore we deem necessary to evaluate them also numerically for a reference fiber as an example. The fiber we consider in the following is step-index with core and cladding refractive indices equal to 1.4660 and 1.4585, respectively (at 1550 nm) and radius $a = 8 \mu\text{m}$. In the C band this fiber supports the propagation of the modes $\text{LP}_{0,1}$, $\text{LP}_{1,1}$ (cut-off at about 3096 nm), $\text{LP}_{2,1}$ and $\text{LP}_{0,2}$ (both with cut-off at about 1943 nm). Regarding mode coupling, we choose the coupling strength using the case of single-mode fibers as a guideline. Specifically, the analysis of the coupling is performed assuming a stress birefringence $\Delta n = 10^{-6}$ (defined as the difference between the refractive indices of the optical axes) and an ellipticity of $\delta a/a = 5 \times 10^{-2}$, where δa is the maximum radius variation [31]. With respect to single-mode fibers, these values correspond to beat lengths of about 1.56 and 3.14 m, respectively. Note however that, as clarified in the following, the coupling coefficients scale linearly with Δn and δa , so the choice of specific values for Δn and δa does not affect the generality of the analysis.

A. From Hybrid Modes to LP Modes

In order to analyze intra-modal dispersion, we must resort to hybrid modes. Nonetheless, mode coupling analysis is more conveniently performed with respect to the simpler LP modes [23], [24]. This is not however a problem, because in the weak-guide approximation LP modes are a linear combination of hybrid modes. The exact expression of this combination is not unique and depends on the chosen expression for the modes and the way in which the modes are sorted.

We refer to the classic expressions of $\text{LP}_{n,p}$ modes reported in Appendix A for completeness. For a given order, we sort the LP modes alternating first x - and y -polarizations and then “even” (cosine) and “odd” (sine) choices, so the sequence is $\text{LP}^{(x,e)}$, $\text{LP}^{(y,e)}$, $\text{LP}^{(x,o)}$, $\text{LP}^{(y,o)}$. Of course the odd choice does not exist when the azimuthal order is zero ($n = 0$).

Under the weak-guide approximation the four-fold degenerate group of $\text{LP}_{n,p}$ modes, for $n \geq 2$, is a linear combination of the four hybrid modes $\text{HE}_{n+1,p}^{(e)}$, $\text{HE}_{n+1,p}^{(o)}$, $\text{EH}_{n-1,p}^{(e)}$ and $\text{EH}_{n-1,p}^{(o)}$ [16], [33]. With the hybrid modes sorted in this order, the transformation matrix from hybrid to LP modes reads [33]

$$\mathbf{T} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & -1 & 0 \end{pmatrix}; \quad (7)$$

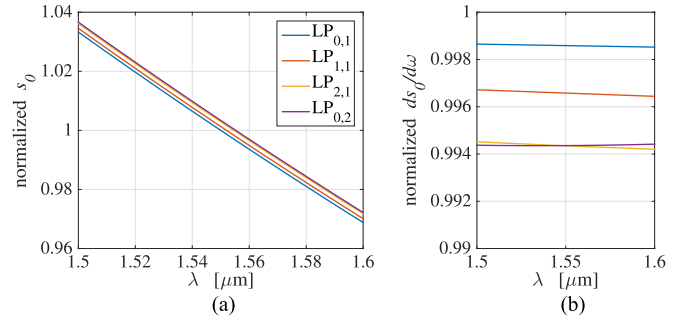


Fig. 1. Coefficient of stress-induced mode coupling, s_0 , versus wavelength for the modes listed in the legend. (a) s_0 normalized to its value at 1550 nm for mode $\text{LP}_{0,1}$. (b) $ds_0/d\omega$ normalized to the value of s_0/ω .

this means that if $\bar{a}$ is the vector of the complex amplitudes of the four hybrid modes, the complex amplitudes of the corresponding LP modes are given by the vector $\mathbf{T}\bar{a}$.

The same transformation matrix holds also for the four-fold degenerate group of $\text{LP}_{1,p}$ modes, provided that $\text{EH}_{0,p}^{(e)}$ and $\text{EH}_{0,p}^{(o)}$ are interpreted as the $\text{TM}_{0,p}$ and $\text{TE}_{0,p}$ modes, respectively. Differently, for the two-fold degenerate group of $\text{LP}_{0,p}$ modes the transformation matrix is simply the identity matrix, since there is a one-to-one correspondence between $\text{LP}_{0,p}^{(x)}$ and $\text{HE}_{1,p}^{(e)}$ and between $\text{LP}_{0,p}^{(y)}$ and $\text{HE}_{1,p}^{(o)}$.

In conclusion, if $\mathbf{K}(z)$ is the coupling matrix evaluated with respect to the $\text{LP}_{n,p}$ group at a given position z , the coupling matrix $\mathbf{C}(z)$ for the corresponding hybrid modes reads (note that $\mathbf{T}$ is orthogonal, thus $\mathbf{T}^{-1} = \mathbf{T}^T$)

$$\mathbf{C}(z) = \mathbf{T}^T \mathbf{K}(z) \mathbf{T}. \quad (8)$$

B. Stress Birefringence

As stated at the beginning of this Section, stress birefringence is one of the two possible sources of intrinsic perturbation. Here we assume that the birefringence is parallel to the reference frame with respect to which the LP modes are defined; we will generalize the results in Section III-D. With this assumption and with respect to $\text{LP}_{n,p}$ modes, the effect of stress birefringence is just a detuning of the x - and y -polarized modes, so the coupling matrix is diagonal and reads $\mathbf{S} = s_0 \text{diag}(1, -1)$, for $n = 0$, and $\mathbf{S} = s_0 \text{diag}(1, -1, 1, -1)$, for $n > 0$, where [23]

$$s_0 = \Delta n \frac{n_{\text{av}} k_0^2}{2\beta_{n,p}}, \quad (9)$$

Δn and n_{av} are the difference and the average between the refractive indices of the two optical axes, respectively, k_0 is the phase constant of vacuum, and $\beta_{n,p}$ is the propagation constant of the corresponding $\text{LP}_{n,p}$ mode. We remark that the coupling matrix due to stress birefringence is the same for all LP modes, except for the dependence on $\beta_{n,p}$.

Fig. 1(a) shows the coefficient s_0 as a function of wavelength for the reference fiber described above and for the different modes listed in the legend. Values are normalized to the value that the same coefficient has at 1550 nm for the fundamental mode $\text{LP}_{0,1}$. As we see, the effect of stress birefringence is practically the same for all the modes. Fig. 1(b) shows $ds_0/d\omega$

normalized to s_0/ω ; the fact that values are extremely close to 1 prove that the approximation $d\mathbf{S}/d\omega \simeq \mathbf{S}/\omega$ is valid in this case.

C. Geometric Deformations

The coupling caused by geometrical deformations is more complex and depends on the considered mode. Assuming that the core is slightly elliptical, one of the main difference is that coupling among $LP_{1,p}$ modes occurs also through the transverse components of the field, whereas all the other LP groups couple only through the longitudinal components. As a result, for a given core ellipticity, the coupling is much stronger (more than two orders of magnitude) for the $LP_{1,p}$ modes [23].

The expressions of the coupling coefficients are rather involved, therefore here we report only the structure of the coupling matrices in terms of some coefficients, g_h , whose expressions are reported in Appendix B for completeness. Note, however, that the coupling coefficients are simply proportional to $\delta a/a$.

The coupling matrix for $LP_{0,p}$ reads $\mathbf{G} = (\delta a/a)g_0 \text{diag}(1, -1)$ and, similarly, the coupling matrix for $LP_{n,p}$, for $n \geq 3$, reads $\mathbf{G} = (\delta a/a)g_1 \text{diag}(1, -1, 1, -1)$; note that both matrices are formally equivalent to those describing stress-induced coupling. Differently, the coupling matrix for $LP_{1,p}$ modes reads [23]

$$\mathbf{G} = \frac{\delta a}{a} \begin{pmatrix} g_3 & 0 & 0 & 0 \\ 0 & g_2 & 0 & 0 \\ 0 & 0 & -g_2 & 0 \\ 0 & 0 & 0 & -g_3 \end{pmatrix} \quad (10)$$

where the difference between g_2 and g_3 is due only to the longitudinal components of the field, so it is negligible. Finally, the coupling matrix for $LP_{2,p}$ modes reads (we remark that there is an error in the analogous matrix reported in Ref. [23])

$$\mathbf{G} = \frac{\delta a}{a} \begin{pmatrix} g_4 & 0 & 0 & g_6 \\ 0 & -g_4 & g_6 & 0 \\ 0 & g_6 & g_5 & 0 \\ g_6 & 0 & 0 & -g_5 \end{pmatrix}. \quad (11)$$

Fig. 2(a) and (c) shows the coefficients g_h as a function of wavelength for the reference fiber described above. Values are normalized to the value that the coefficient g_0 has at 1550 nm for the fundamental mode $LP_{0,1}$. Differently from the case of stress birefringence, the effect of core ellipticity strongly depends on the modes. In particular and as anticipated, the coupling coefficients of modes $LP_{1,p}$ [see Fig. 2(c)] are much larger than those for the order modes; note also that the difference between g_2 and g_3 is negligible. Fig. 2(b) and (d) shows $dg_h/d\omega$ normalized to g_h/ω for the same mode. Again at variance with stress birefringence, in this case the ratios are quite different from 1, meaning that for geometric deformations the approximation $d\mathbf{G}/d\omega \simeq \mathbf{G}/\omega$ is not accurate.

D. Perturbation Orientation

The matrix expressions reported above are based on the assumption that the perturbation is oriented parallel to the reference frame with respect to which the LP modes are defined. So for example, the two main axes of the elliptical core are

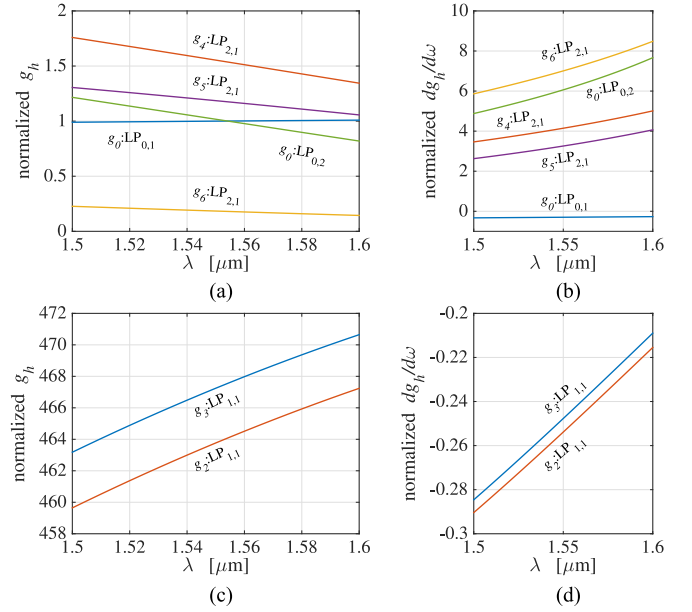


Fig. 2. Mode coupling induced by stress birefringence versus wavelength. (a), (c) Coefficients g_h for the modes as indicated in the graphs, normalized to the value of g_0 at 1550 nm for mode $LP_{0,1}$. (b), (d) Derivatives $dg_h/d\omega$ normalized to g_h/ω .

parallel to the x - and y -polarizations of modes. To generalize the results we should express them with respect to an arbitrary reference frame. Mathematically this means projecting the LP modes parallel to the perturbation on a set of modes with arbitrary orientation. After cumbersome calculations it can be shown that this projection is implemented by the following rotation matrices [23]:

$$\mathbf{R}_0(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad \text{for } LP_{0,p} \quad (12)$$

$$\mathbf{R}_n(\theta) = \mathbf{R}_{\text{pol}}(\theta)\mathbf{R}_{\text{pat}}(n\theta), \quad \text{for } LP_{n,p}, \quad n \geq 1, \quad (13)$$

where $(\mathbf{R}_{\text{pol}})$ is defined by 2×2 blocks)

$$\mathbf{R}_{\text{pol}}(\theta) = \begin{pmatrix} \mathbf{R}_0(\theta) & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_0(\theta) \end{pmatrix}, \quad (14)$$

$$\mathbf{R}_{\text{pat}}(n\theta) = \begin{pmatrix} \cos n\theta & 0 & -\sin n\theta & 0 \\ 0 & \cos n\theta & 0 & -\sin n\theta \\ \sin n\theta & 0 & \cos n\theta & 0 \\ 0 & \sin n\theta & 0 & \cos n\theta \end{pmatrix}. \quad (15)$$

The mathematical details apart, it is known that $\mathbf{R}_0$ represents a rotation of polarization; similarly, $\mathbf{R}_{\text{pol}}$ represents a rotation of polarization within the “odd” and “even” couples of LP modes. Differently, $\mathbf{R}_{\text{pat}}$ describes a rotation of the field pattern, which is why it depends on the mode azimuthal order n and it acts on each polarization independently. We remark also that $\mathbf{R}_{\text{pol}}$ and $\mathbf{R}_{\text{pat}}$ commute and that $\mathbf{R}_n^{-1} = \mathbf{R}_n^T$.

The above analysis allows us to conclude that if $\mathbf{K}'$ is the coupling matrix for the $LP_{n,p}$ modes and for a perturbation aligned with the reference axis, the coupling matrix of the same perturbation if rotated by an angle θ (positive for counterclockwise rotation) reads:

$$\mathbf{K}(\theta) = \mathbf{R}_n^T(\theta)\mathbf{K}'\mathbf{R}_n(\theta). \quad (16)$$

E. Superposition of Perturbations

In the limits of validity of coupled mode theory, the combined effect of more perturbations is described by the sum of the coupling matrices of each perturbation. In general each perturbation may have a different orientation, so the respective matrices should be transformed according to (16) before being summed.

For what concerns intrinsic perturbations, we may reasonably assume that stress birefringence and geometric deformations are aligned, because indeed an elliptical core originates a stress birefringence aligned with the axes of the ellipsis [31]. On the other hand, however, stress birefringence might be induced also by cladding asymmetries and this may cause a misalignment with the core geometry; nonetheless we think that this effect can be neglected at this stage. Accordingly, we write the coupling matrix of intrinsic perturbation for the arbitrary $LP_{n,p}$ modes as

$$\mathbf{K}(\theta) = \mathbf{R}_n^T(\theta)(\mathbf{S} + \mathbf{G})\mathbf{R}_n(\theta), \quad (17)$$

where θ is the angle of orientation of the perturbation.

Regarding the strength of the two sources of perturbation, previous analyses performed on single-mode fibers [31] have shown that they may have slight dependence on the index profile. For example, the analysis reported in Ref. [31] shows that the ratio g_0/s_0 depends on the index profile, and varies from about 1 for step-index fiber to about 0.95 for non-zero dispersion fibers. This ratio may likely vary for multi-mode fibers, owing to the larger core and different stress distribution; however, we are not aware of any work analyzing the relationship between stress birefringence and core ellipticity in multi-mode fibers. Therefore, in Section V we numerically analyze the two sources of coupling independently for varying strength; actually, we have numerically verified that the differences between the combined action of the two perturbations and the action of each of them independently are negligible, as long as the overall mode coupling strength, $\Delta\kappa$, is kept constant.

IV. MODELING SPUN FIBERS

Spin consists in rotating the fiber during the drawing process. We assume that the only effect of spin is to rotate the fiber and its intrinsic perturbations, without applying any torsional stress or affecting the strength of the intrinsic coupling. Regarding the absence of torsional stress, the assumption seems reasonable because the spin is applied when the silica is still in a molten state. Differently, there are experimental evidences that in single-mode step-index fibers the spin may slightly influence the strength of mode coupling by relieving some of the stress in the core. [34] Something similar might happen also for multi-mode step-index fibers, but at this stage we neglect this effect for simplicity. Therefore, the coupling matrix of a spun fiber with respect to the $LP_{n,p}$ modes can be written as

$$\mathbf{K}_n(z) = \mathbf{R}_n^T(\theta + \alpha)(\mathbf{S} + \mathbf{G})\mathbf{R}_n(\theta + \alpha), \quad (18)$$

where $\alpha = \alpha(z)$ is the applied spin, i.e., the angle by which the fiber is rotated at a given distance z , θ is the orientation

of the intrinsic perturbations, and the perturbation strength is represented by the matrices $\mathbf{S}$ and $\mathbf{G}$.

The spin profile, $\alpha(z)$, can be an arbitrary function, but periodic ones are usually preferred for practical reasons. In this work we consider sinusoidal spin, $\alpha(z) = A \sin(2\pi z/P)$, where A is the spin amplitude and P the spin period.

In general also θ , $\mathbf{S}$ and $\mathbf{G}$ are function of z , because intrinsic perturbations vary at random along the fiber. Almost all of the analyses of modal dispersion published so far have described mode coupling variations in terms of white noises [22], [27], [35], [36]. This choice, driven only by mathematical convenience, is however not suitable to study the spin. Actually, the spin is a deterministic continuous rotation superposed to the intrinsic rotation of perturbation orientation. If this intrinsic orientation were varying as a white noise, any effect of the spin would be just blurred away. In other words, to study the spin we need to take into account the fact that the orientation of the intrinsic perturbation in real fibers has a finite correlation length. To this aim, we describe the angle θ with a Wiener process [30], i.e., we assume that

$$\frac{d\theta}{dz} = -\sigma\eta(z), \quad (19)$$

where $\eta(z)$ is a Gaussian white noise, such that $\langle \eta(z)\eta(u - z) \rangle = \delta(u)$, and the parameter σ influences the correlation length of the fiber, L_F , by the relation $L_F = 1/\sigma$ [30].

In principle also perturbation strength may vary randomly along the fiber; however, in analogy with the so called fixed-modulus model used in PMD analysis [30], here we assume for simplicity that perturbation strength (i.e., the coefficients s_0 and g_h) is constant along the fiber. Studies on single-mode spun fibers have shown that a fixed perturbation strength tends to sharpen spin effects, deepening and narrowing minima of the SRF. On the contrary, a more realistic randomly varying perturbation strength would blur those minima toward a sort of average behavior [37]. This effect is likely to occur also in multi-mode fibers and has to be kept in mind while analyzing numerical results.

We can now cast the complete model to analyze intra-modal dispersion in spun fibers. With respect to the Eq. (6) and with respect to the hybrid modes that make the generic $LP_{n,p}$ group, using (7) the matrix $\mathbf{B}$ can be written as:

$$\mathbf{B}(z) = \mathbf{D} + \mathbf{T}^T \mathbf{R}_n^T(\theta + \alpha)(\mathbf{S} + \mathbf{G})\mathbf{R}_n(\theta + \alpha)\mathbf{T}, \quad (20)$$

where we recall that by the above assumptions the only quantities depending on z are the perturbation intrinsic orientation $\theta(z)$, governed by (19), and the spin profile $\alpha(z)$.

The ω -derivative of $\mathbf{B}$ depends on the dependence on ω of $\mathbf{D}$, $\mathbf{S}$ and $\mathbf{G}$. As anticipated, we assume for simplicity that $d\mathbf{B}/d\omega = \mathbf{B}/\omega$, but we perform also control simulations with the exact values of $d\mathbf{B}/d\omega$.

A. Five Main Cases

The mathematical analogies among the coupling matrices $\mathbf{S}$ and $\mathbf{G}$ for different modes allow us to single out five main cases that are representative of all the other modes (see Fig. 3).

	LP(0,p)	LP(1,p)	LP(2,p)	LP(n≥2,p)
Birefringence	LP(0,1) & birefringence	LP(1,1) & birefringence	LP(2,1) & birefringence	
Core Ellipticity		LP(1,1) & core ellipticity	LP(2,1) & core ellipticity	

Fig. 3. Schematic representation of the five main cases in which different modes and different intrinsic perturbations can be cast.

First of all notice that the radial order, p , of the modes does not affect the mathematical expressions encountered so far, as it only influences the magnitude of the parameters. This let us conclude that spin will have the same effects on two modes with the same azimuthal order, n , but different radial order, provided that the respective parameters are rescaled.

Let us now consider the $LP_{0,p}$ modes. As shown in Sections III-B and III-C, the corresponding coupling matrices are mathematically the same, both for stress-birefringence and core ellipticity. Furthermore, $LP_{0,p}$ modes are exactly degenerate, so $\mathbf{D} = 0$ in (20). Consequently, the effects of spin on these modes is exactly equivalent to the effects on the fundamental mode $LP_{0,1}$, which are already widely documented in the literature [13], [38]–[40]. We can therefore straightforwardly extend the results obtained for $LP_{0,1}$ to $LP_{0,p}$ modes, regardless of the kind of intrinsic perturbation. We recall the main results for completeness [13]:

- 1) In general the spin effectiveness in reducing the SIRF increases with the spin-rate.
- 2) The relationship between SIRF and spin-rate for periodic spin function is not monotonic; optimal spin amplitudes exist for which the SIRF has local minima.
- 3) The effects of spin are the same independently of whether the perturbation strength is fixed or varying, provided that the spin period is shorter than the coupling beat-length.
- 4) If the perturbation variation along the fiber is sufficiently smooth, and the spin profile has a particular asymmetry (something however not usually encountered in practice), the spin may actually increase the DGD rather than reduce it [15].

Next, notice that the coupling matrix due to core ellipticity for $LP_{n,p}$ with $n \geq 3$ is mathematically equivalent to that due to stress birefringence for $LP_{n,p}$ with $n \geq 1$. However, modes $LP_{1,p}$ are made of three non-degenerate hybrid modes, differently from the others that are made of two couples of two-fold degenerate modes. This means that the $\mathbf{D}$ matrix for the $LP_{1,p}$ is not mathematically equivalent to the $\mathbf{D}$ matrix for modes with $n \geq 2$. As a consequence, with respect to spin, the $LP_{2,1}$ mode with stress birefringence is mathematically equivalent to (i) $LP_{n,p}$ modes with $n \geq 2$ and stress birefringence and (ii) $LP_{n,p}$ modes with $n \geq 3$ and core ellipticity (see Fig. 3).

Differently modes $LP_{2,p}$ with core ellipticity and modes $LP_{1,p}$ with either stress-birefringence or core ellipticity constitute three cases on their own, represented by the respective mode with radial order $p = 1$ and the respective perturbation.

For the sake of completeness, we remark that the mathematical analogies discussed so far regard matrix $\mathbf{B}$, but should be

extended also to $d\mathbf{B}/d\omega$. This is automatically achieved under the usual assumption $d\mathbf{B}/d\omega \simeq \mathbf{B}/\omega$, but we already said that this assumption is not always verified. The cases we have just identified are therefore most suited for a qualitative analysis of intra-modal dispersion, which we still deem worth of interest. The first class, $LP_{0,1}$, has been already widely studied. The other four are numerically analyzed in the following section.

V. NUMERICAL RESULTS

The four main cases relative to $LP_{n,p}$ modes with $n \geq 1$ have been numerically simulated for different parameters. In order to compare the different cases, we perform the analysis for the same coupling strength, $\Delta\kappa$, and the same modal birefringence, $\Delta\beta$. However, fixing only $\Delta\beta$ and $\Delta\kappa$ does not allow to define completely the matrices $\mathbf{D}$, $\mathbf{S}$ and $\mathbf{G}$, since the mutual ratios of their elements would be left undetermined. Therefore, first we calculate the propagation constants and coupling coefficients for the reference fiber described at the beginning of Section III. Then, we linearly rescale by a proper factor the matrices $\mathbf{D}$, $\mathbf{S}$ and $\mathbf{G}$, so to achieve the chosen values of $\Delta\beta$ and $\Delta\kappa$.

Simulations are performed for sinusoidal spin with period $P = 4$ m and amplitudes, A , ranging from 0 to 6 rad. Stress birefringence and core ellipticity are simulated separately, with coupling strength $\Delta\kappa \simeq 0.42$ rad/m (corresponding to a coupling beat length of 15 m) in both cases. Modal birefringence $\Delta\beta$ is varied from 0 (the case in which LP modes are exactly degenerate) to about 21 rad/m (corresponding to a mode beat length of 30 cm). The fiber length is 1 km and perturbation orientation is varied randomly along the fiber according to (19), with fiber correlation length $L_F = 10$ m [41]. The statistical ensemble is made of 2000 fiber realizations for each set of parameters.

The transmission matrix of the fiber $\mathbf{U}$ has been calculated using the waveplate model [42], [43] as $\mathbf{U} = \mathbf{W}_N \mathbf{W}_{N-1} \dots \mathbf{W}_1$, where $\mathbf{W}_m = \exp[-j(\mathbf{B}_m)L_S]$ is the transmission matrix of the m th waveplate and L_S is its length (set to 5 cm in simulations). Matrix $\mathbf{U}$ is calculated at two sufficiently close frequencies ω and $\omega + \delta\omega$, so that mode delays, $\tau_i(z)$, can be calculated as $\tau_i = \arg(\mu_i)/\delta\omega$, where $\mu_i = \mu_i(z)$ are the eigenvalues of the matrix $\mathbf{U}^*(z, \omega + \delta\omega)\mathbf{U}(z, \omega)$ [21].

A. Results for Approximate Frequency Dependence

The results of numerical simulations assuming $d\mathbf{B}/d\omega = \mathbf{B}/\omega$ are shown in Fig. 4, where the SIRF for the four main cases as indicated in each graph is plotted as a function of spin amplitude A and modal birefringence $\Delta\beta$ (note that the horizontal axes are nonlinearly scaled to highlight some features). The SIRF is encoded in colors; in particular white zones correspond to $\text{SIRF} = 1$. Several properties of the spin can be grasped from those graphs.

First of all notice that results are qualitatively the same in all cases, although a difference is visible between $LP_{1,1}$ and $LP_{2,1}$ modes. On the contrary, for a given mode, the SIRF is almost the same for both kinds of perturbation; so, despite the coupling matrices are different, the effect of stress birefringence and core ellipticity is similar. As a consequence, results do not change

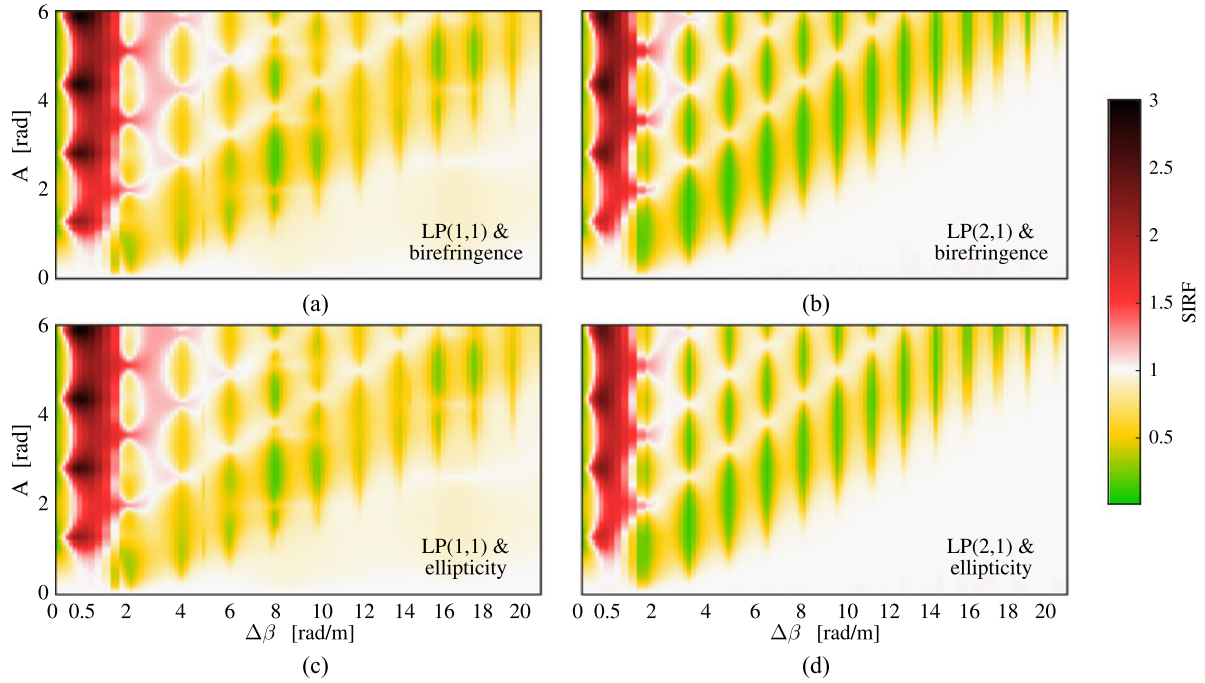


Fig. 4. SIRC as a function of modal birefringence, $\Delta\beta$, and spin amplitude, A , for the four main cases as indicated in each graph. Graphs share the same axes; the horizontal axes are nonlinearly scaled to highlight some features for low values of $\Delta\beta$.

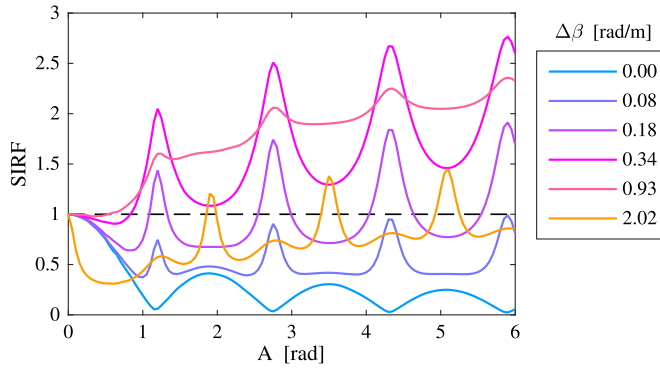


Fig. 5. SIRC as a function of spin amplitude, A , for different values of modal birefringence, $\Delta\beta$, as indicated in the legend, and for the case of $LP_{1,1}$ with birefringence.

even when the two perturbations are acting at the same time, provided that the overall coupling strength, $\Delta\kappa$, is the same.

One of the most striking feature is that, as anticipated, there is a range of the parameters where the SIRC is largely above 1; in this region the spin is increasing the DGD, rather than reducing it. This is more evident in Fig. 5, where the SIRC is plotted as a function of A for different values of $\Delta\beta$, for $LP_{1,1}$ and birefringence. While we do not yet have a theoretical explanation for these odd results, we can provide a reasonable explanation.

Start considering that the two fundamental $LP_{0,1}$ modes of an ideal single-mode fiber are degenerate, therefore the ideal fiber has no modal dispersion. Random mode coupling, however, breaks this degeneracy causing modal dispersion (i.e., PMD) to grow; so in a single-mode fiber random mode coupling is a detrimental effect. When the spin is applied, it mitigates the

effects of random mode coupling by recovering “on average” the perfect cylindrical symmetry of the ideal fiber [14]. Differently, in an ideal multi-mode fiber, the group of $LP_{n,p}$ modes with $n \geq 1$ is actually made of non-degenerate hybrid modes, therefore it is already affected by some modal dispersion. In the ideal fiber this dispersion grows linearly with distance, whereas in real fibers it grows on average with the square root of distance, just because of the random mode coupling taking place along the fiber. So in multi-mode fibers random mode coupling is a beneficial effect. But just as in single-mode fibers, also in multi-mode fibers spin tends to mitigate the effects of random mode coupling, leaving however the intrinsic modal dispersion unopposed. As a result, the more effective is the spin in opposing random mode coupling, the more the DGD increases.

The effectiveness of this mechanism depends, however, on the strength of the interacting phenomena. In the somewhat ideal case in which modes $LP_{n,p}$ ($n \geq 1$) are exactly degenerate ($\Delta\beta = 0$), the spin is quite effective in reducing DGD just as in single-mode fibers (see Fig. 5). Nonetheless, as the modal birefringence increases, the spin starts losing effectiveness and the SIRC tends to go above 1; this is more evident just for those amplitudes where the spin is most effective when $\Delta\beta = 0$. Quite interestingly, however, the region where the SIRC is largely above 1 occurs only for a small range of modal birefringence, which is around the same value of the coupling strength $\Delta\kappa$ (0.42 rad/m in the present simulations). Beyond that region, for larger values of $\Delta\beta$, the spin returns to be effective in reducing the DGD, and the SIRC has several local minima that occurs with a certain periodicity.

Therefore, when modal birefringence and mode coupling have comparable strengths, the spin may actually disrupt the beneficial effects that random mode coupling has on modal

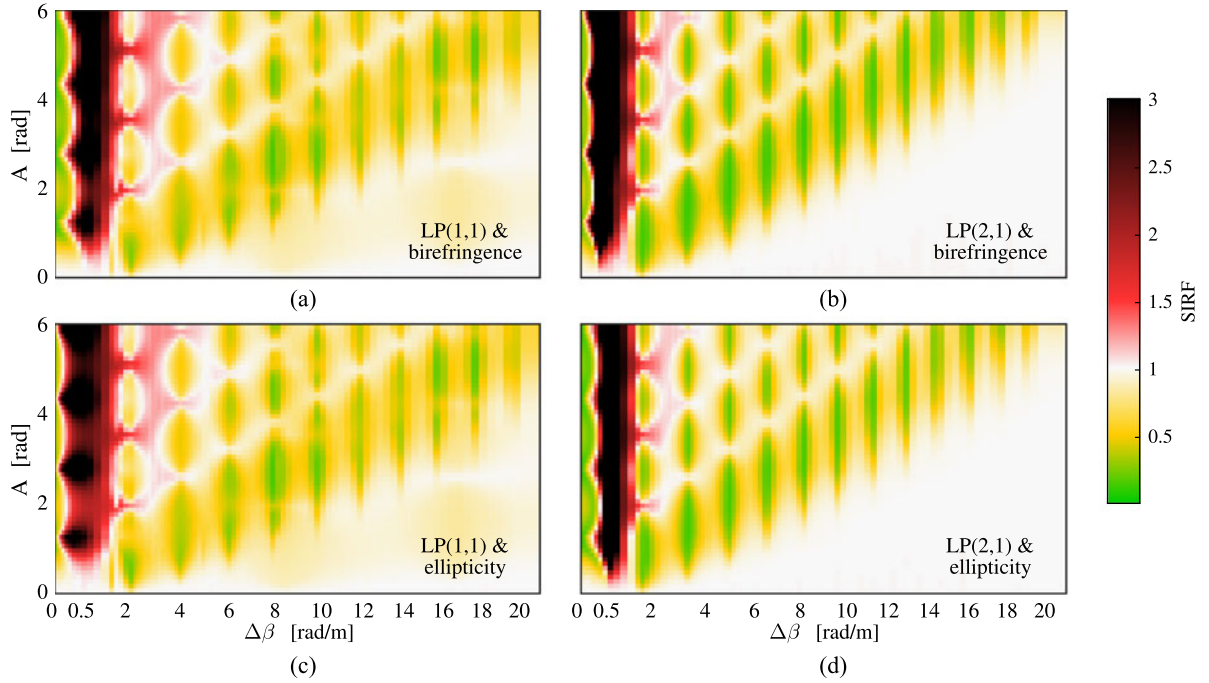


Fig. 6. SIRC calculated with the exact frequency dependence of $\mathbf{B}(\omega)$ as a function of modal birefringence, $\Delta\beta$, and spin amplitude, A , for the four main cases as indicated in each graph. Graphs share the same axes; the horizontal axes are nonlinearly scaled to highlight some features for low values of $\Delta\beta$. The wide black areas are due to the fact that the SIRC goes well above 3.

dispersion. Differently, when the modal birefringence is larger than mode coupling the spin is still effective in reducing DGD, although the effectiveness is on average higher for $LP_{n,p}$ modes with $n \geq 2$ than for $LP_{1,p}$.

We remark that this effect is very specific to modes with azimuthal order $n \geq 1$, while it does not occur for $LP_{0,p}$ modes. Actually the effect arises from the interplay between spin, mode coupling and modal birefringence, but this last term is intrinsically zero for $LP_{0,p}$ modes. Moreover, the present effect should not be confused with the effect mentioned in point 4 of Section IV-A, which occurs under quite different and uncommon conditions not analyzed here.

Finally, another feature shown in Fig. 4 is the large triangular white areas in the right lower corners of the graphs. Those areas indicate that the spin can reduce the DGD only if the spin-rate is fast enough compared to the modal birefringence.

B. Results for Exact Frequency Dependence

As anticipated, the numerical analyses have been performed also considering the exact dependence of $\mathbf{B}$ on ω . As shown in Fig. 6, the exact model is in general comparable with the approximate one (see Fig. 4), but a marked difference occurs just for the values of $\Delta\beta$ where the SIRC is larger than 1.

The same data are shown more in detail in Fig. 7(a), where the SIRC is plotted as a function of modal birefringence, $\Delta\beta$, for different values of spin amplitude, A , and for $LP_{2,1}$ mode and birefringence. Continuous curves refer to the exact model, whereas dashed ones refer to the approximate model (see Fig. 4). As we see, the two kinds of curve are almost overlapped, except for the range of $\Delta\beta$ from 0 to about 1.8 rad/m, where the SIRC of the exact model reaches quite unusually high values [see

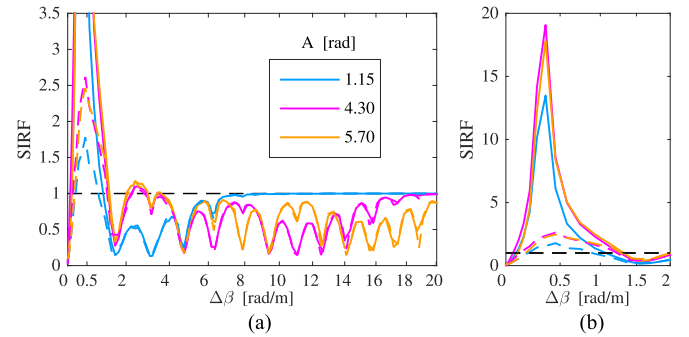


Fig. 7. (a) SIRC as a function of modal birefringence, $\Delta\beta$, for different values of spin amplitude, A , as indicated in the legend, and for $LP_{2,1}$ and birefringence. Continuous and dashed curves refer to the exact and approximate frequency dependence, respectively. (b) Detail of graph (a).

Fig. 7(b)]. While we believe that these results are specific to the case we have simulated, they indeed confirm that spin is effective only when the modal birefringence of the fiber is not comparable with its mode coupling strength.

Finally notice also that with the exact model the SIRC has a small dependence on the kind of perturbation, not seen with the approximate model. This difference, however, regards only the $LP_{1,1}$ mode again for the range of $\Delta\beta$ values where the spin is larger than 1 [compare Fig. 6(a) and (c)].

C. Comparison Between SIRC and $SIRC_{rms}$

We conclude the analysis by comparing the SIRC, as defined in (4) with respect to $\langle\Delta\tau\rangle$, and the $SIRC_{rms}$, as defined in (5) with respect to $\langle\tau^2\rangle$. We recall that the interest in $SIRC_{rms}$ is just due to the fact that it is likely easier to analytically evaluate $\langle\tau^2\rangle$

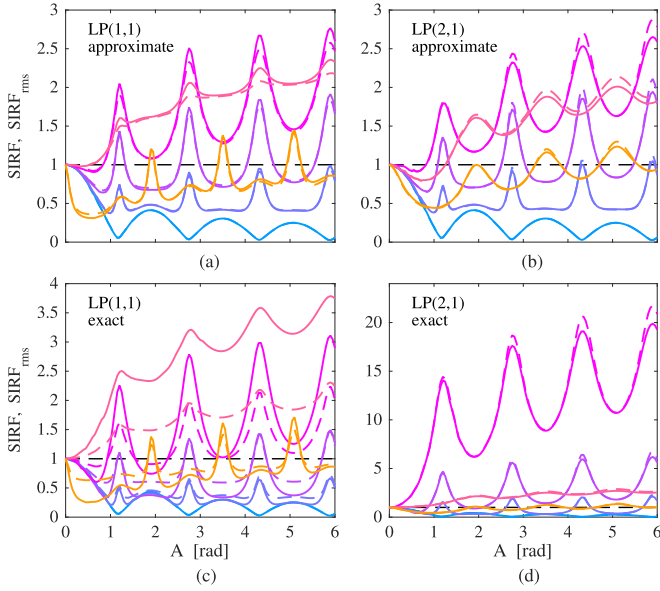


Fig. 8. SIRF (continuous curves) and SIRF_{rms} (dashed curves) as a function of spin amplitude, A , for different values of modal birefringence, $\Delta\beta$, as indicated in the legend of Fig. 5, for different modes and models as indicated in the graphs, and for the case of stress birefringence. Graphs share the same horizontal axis.

rather than $\langle\Delta\tau\rangle$. However, it would be pointless to calculate SIRF_{rms} if it were not related to the SIRF, i.e., the parameter of interest.

Fig. 8 shows a comparison between the results obtained with the two models. For the approximate model [see Fig. 8(a) and (b)] the SIRF (continuous curve) and the SIRF_{rms} (dashed curves) are almost the same, with marginal differences only for SIRF values larger than 1 (a range of low practical interest). The same is true also for the exact model and $\text{LP}_{2,1}$ mode [see Fig. 8(d)], but not for the exact model and $\text{LP}_{1,1}$ mode [see Fig. 8(c)], where marked differences can be seen especially when the SIRF is larger than 1. Nevertheless, the behavior of the two parameters is qualitatively the same. Therefore we may conclude that, at least for the range of parameters with some practical interest, where the SIRF is lower than 1, the SIRF can be accurately evaluated with the approximated model with respect to the rms-MD.

VI. CONCLUSION

In this work we have numerically analyzed the influence of spin on the intra-modal dispersion of step-index multi-mode fibers. The analysis has considered the effects of stress birefringence and geometric deformation of the core, the two main sources of intrinsic mode coupling. With the aid of some analytic considerations, we have reduced the most general problem to five main cases, represented in Fig. 3. The first case includes all the $\text{LP}_{0,p}$ and is mathematically analogous to $\text{LP}_{0,1}$; as a consequence of this analogy, all the results known about single-mode spun fibers can be extended to the $\text{LP}_{0,p}$ modes. The other four cases have been numerically analyzed.

Results have shown that in general spin can reduce the DGD also for high-order modes, but there is a range of parameters

where the spin increases the DGD rather than reducing it. This occurs when the modal birefringence is comparable with random mode coupling, and the effect is most evident just for those spin amplitudes that would give maximum DGD reduction in the case of degenerate modes (e.g., as for $\text{LP}_{0,1}$). We believe that this happens because when modal birefringence and mode coupling are comparable, the random action of the latter is most effective in mitigating the former. In this situation, however, the spin tends to mitigate the effects of mode coupling leaving the modal birefringence unopposed, thus causing an increase of the DGD. Differently, when modal birefringence is weaker than mode coupling, the main source of modal dispersion is mode coupling, which is however mitigated by the spin; therefore the DGD is reduced. Finally, when modal birefringence is stronger than random mode coupling, the mitigating effect of the latter is weak, but the spin contributes causing also in this case a reduction of the DGD.

In general the higher the spin-rate, the stronger the effect of the spin (both in decreasing or increasing the SIRF). There are however some sort of resonances where the spin induces local maxima and minima. These resonances have an almost periodic recurrence with respect to both the spin amplitude, A , and the modal birefringence, $\Delta\beta$.

These results confirm that a properly designed spin is a viable approach in reducing intra-modal dispersion of multi-mode fibers. This may have interesting application in the weakly-coupled approach to SDM, where each set of $\text{LP}_{n,p}$ modes are used as independent transmission channels. It is worthwhile remarking that in this case spin would not compromise the weak-coupling condition, because it would have no effect on the intentionally large inter-modal birefringence of the fibers designed for this application.

We have numerically explored also the effects of an approximate dependence on frequency, showing that the results obtained with the common assumption $d\mathbf{B}/d\omega = \mathbf{B}/\omega$ are comparable with the exact ones in the cases of practical interest. Similarly, also the results about the rms-MD, τ , are comparable with those obtained for the DGD, $\Delta\tau$. Given the analytic simplicity with which τ can be calculated, and given the simplification given by the approximate frequency dependence, the above facts pave the way to the development of a comprehensive theoretical analysis that would ultimately clarify and explain the role of spin in managing and control modal dispersion of multi-mode fibers.

The analysis reported here describes the effects of spin on the intra-modal dispersion of higher-order modes of step-index fibers. As long as coupling among non-degenerate modes can be neglected, the analysis applies also to graded-index fibers, at least qualitatively. On the contrary, the analysis does not consider the effects of spin on strongly-coupled fibers, which is still a completely unexplored topic.

APPENDIX A EXPRESSION OF LP MODES

The mode coupling analysis reported in the text is based on the classical expression of LP modes [16], [33]. Accordingly, the transverse component of the electric field of the $\text{LP}_{n,p}$ mode can be expressed as $\bar{E}_t = F_{n,p}(r)\Psi(n\theta)\hat{e}$, where r and θ are the

radial and azimuthal coordinate on the fiber section, $\hat{e}$ represent field polarization and can be either $\hat{x}$ or $\hat{y}$, $\Psi(n\theta)$ can be either $\cos(n\theta)$ or $\sin(n\theta)$, and

$$F_{n,p}(r) = \begin{cases} \frac{J_n(ur)}{J_n(u)}, & \text{if } r \leq a, \\ \frac{K_n(wr)}{K_n(w)}, & \text{if } r > a, \end{cases} \quad (21)$$

where a is the core radius, J_n is the Bessel function of first kind, K_n is the modified Bessel function of second kind, and

$$u = \sqrt{k_0^2 n_{co}^2 - \beta^2}, \quad w = \sqrt{\beta^2 - k_0^2 n_{cl}^2}, \quad (22)$$

where β is the propagation constant and $k_0 = \omega\sqrt{\epsilon_0\mu_0}$. The longitudinal component can be calculated as

$$E_z = -j(\nabla \cdot \bar{E}_t)/\beta. \quad (23)$$

APPENDIX B

EXPRESSION OF MODE COUPLING COEFFICIENTS

The expression of the coupling coefficients obtained from the analysis of coupled modes are listed below:

$$g_0 = (NA k_0)^2 \frac{[aF'_{0,p}(a)]^2}{8Q_{0,p} \beta_{0,p}^3}, \quad (24)$$

$$g_1 = (NA k_0)^2 \frac{[aF'_{n,p}(a)]^2 - [nF_{n,p}(a)]^2}{8Q_{n,p} \beta_{n,p}^3}, \quad (n \geq 3), \quad (25)$$

$$g_2 = (NA k_0)^2 \frac{a^2 F_{1,p}^2(a)}{4Q_{1,p} \beta_{1,p}}, \quad (26)$$

$$g_3 = g_2 + (NA k_0)^2 \frac{[aF'_{1,p}(a)]^2 - F_{1,p}^2(a)}{4Q_{1,p} \beta_{1,p}^3}, \quad (27)$$

$$g_4 = (NA k_0)^2 [3aF'_{2,p}(a) - 2F_{2,p}(a)] \times \frac{aF'_{2,p}(a) + 2F_{2,p}(a)}{16Q_{2,p} \beta_{2,p}^3}, \quad (28)$$

$$g_5 = (NA k_0)^2 [aF'_{2,p}(a) - 6F_{2,p}(a)] \times \frac{aF'_{2,p}(a) + 2F_{2,p}(a)}{16Q_{2,p} \beta_{2,p}^3}, \quad (29)$$

$$g_6 = (NA k_0)^2 \frac{[aF'_{2,p}(a) + 2F_{2,p}(a)]^2}{16Q_{2,p} \beta_{2,p}^3}, \quad (30)$$

where n_{co} and n_{cl} are the refractive indices of core and cladding, respectively, $NA = \sqrt{n_{co}^2 - n_{cl}^2}$ is the numerical aperture and

$$Q_{n,p} = \int_0^\infty r F_{n,p}^2(r) dr. \quad (31)$$

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